

LightGBM (Standard)

Acer Pollen Prediction – Original Scale

(Duan Smearing Back-transformation)

Pooled model (all stations), 5-seed average, early stopping, Bayesian-optimized

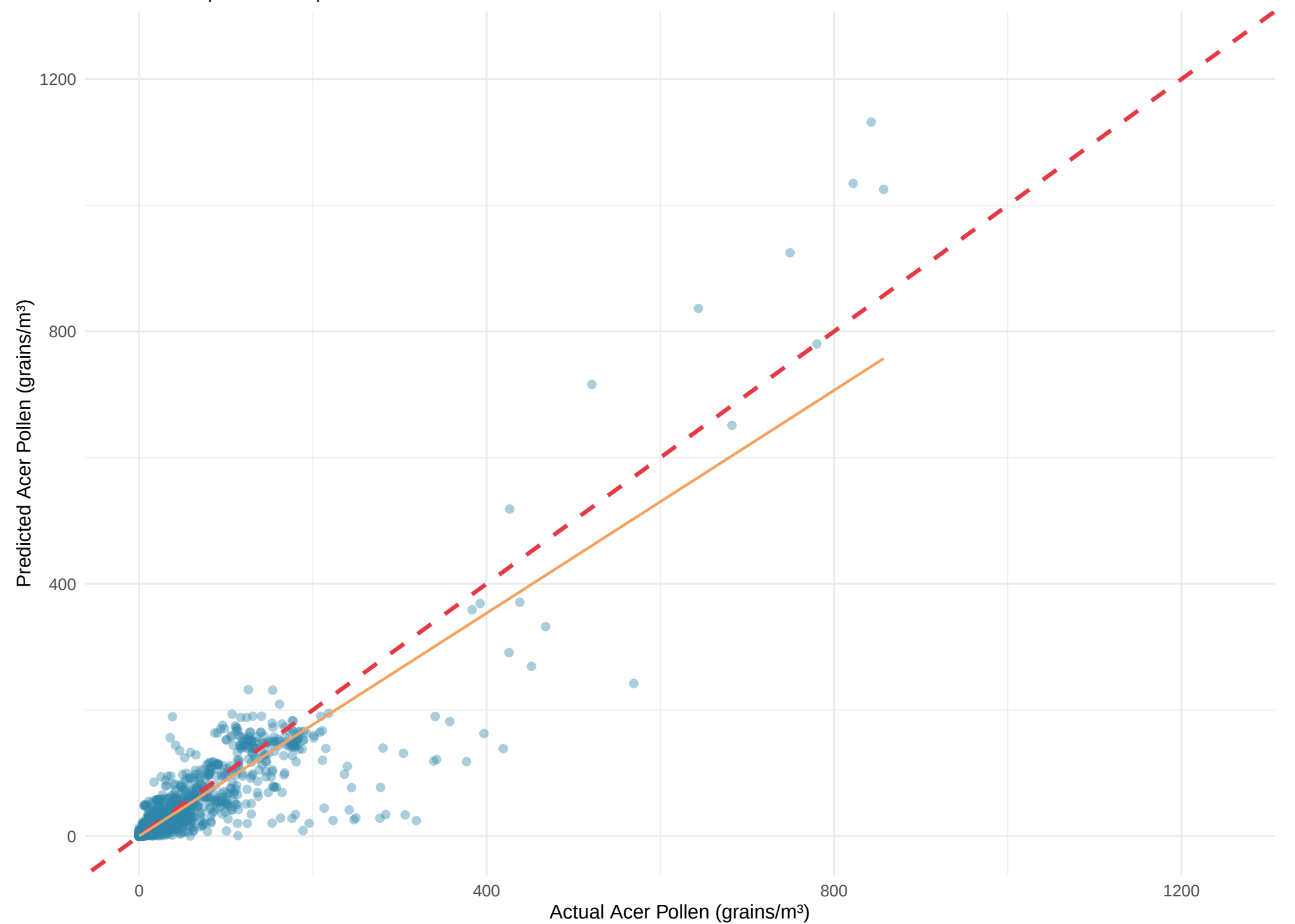
Duan Smearing Factor: 1.0025

Test Performance (Original Scale):

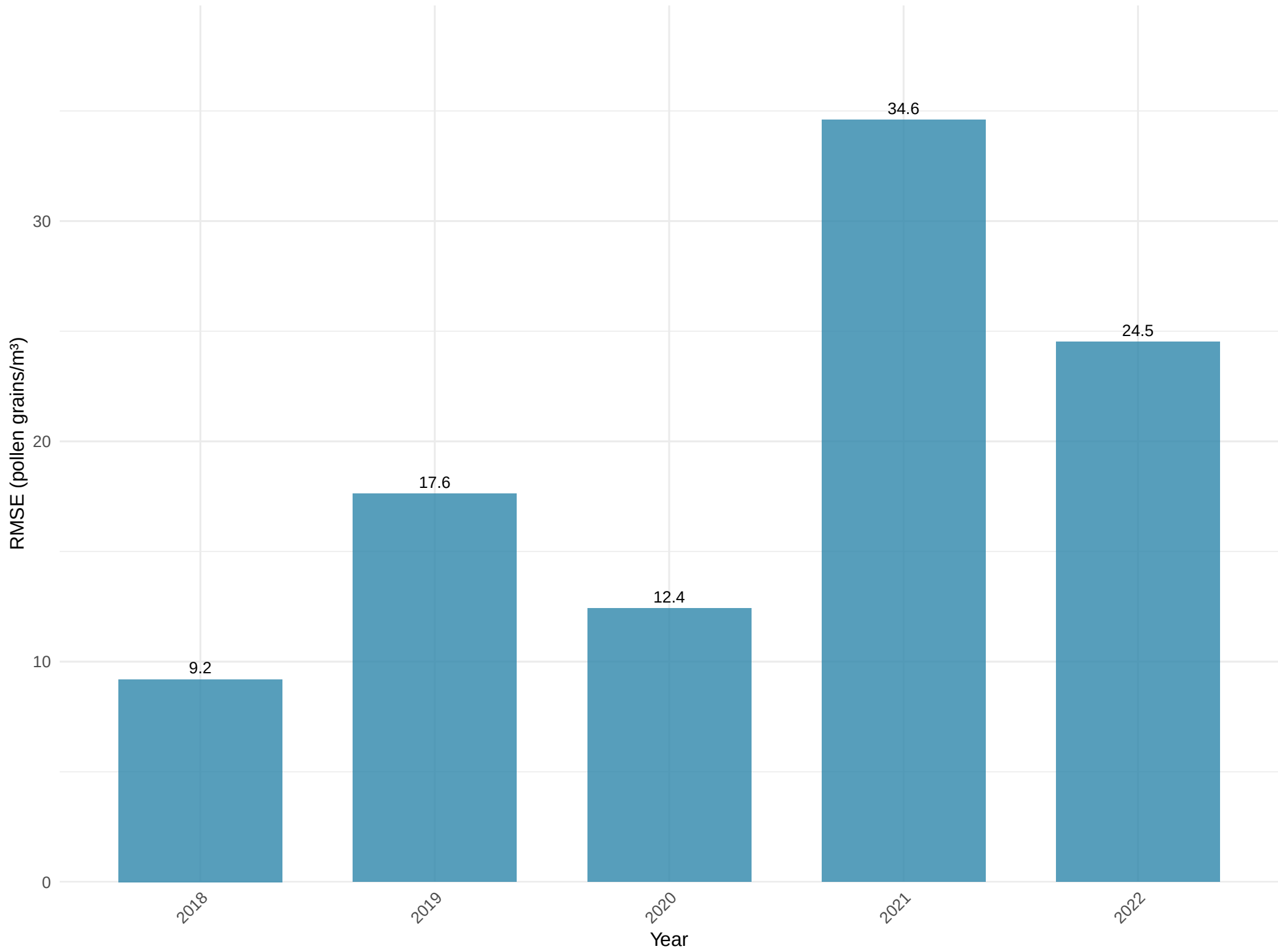
RMSE: 16.34 | MAE: 3.67 | R^2 : 0.8159

Actual vs Predicted (Original Scale)

RMSE: 16.34 | MAE: 3.67 | R²: 0.8159

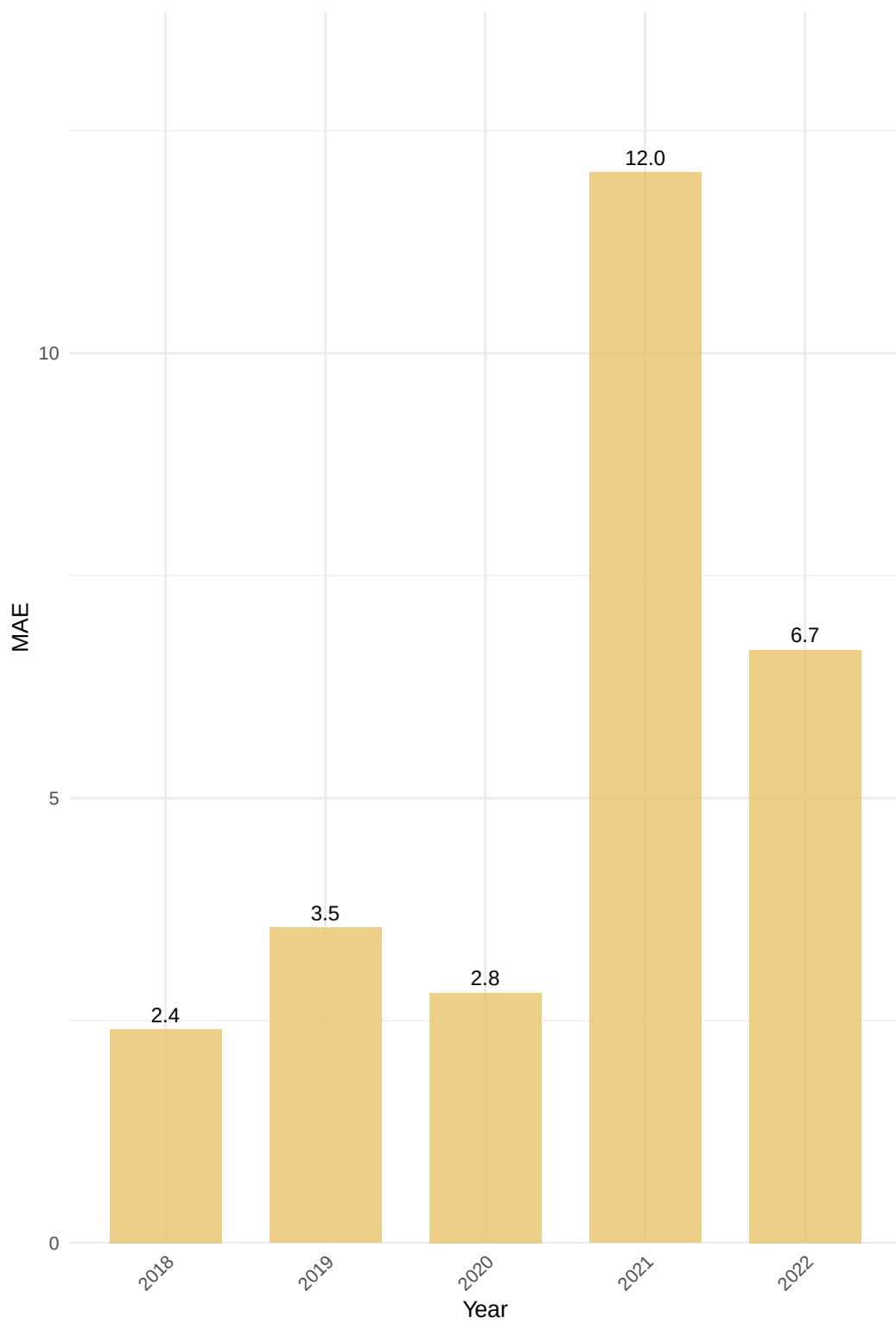


RMSE by Year (Original Scale)

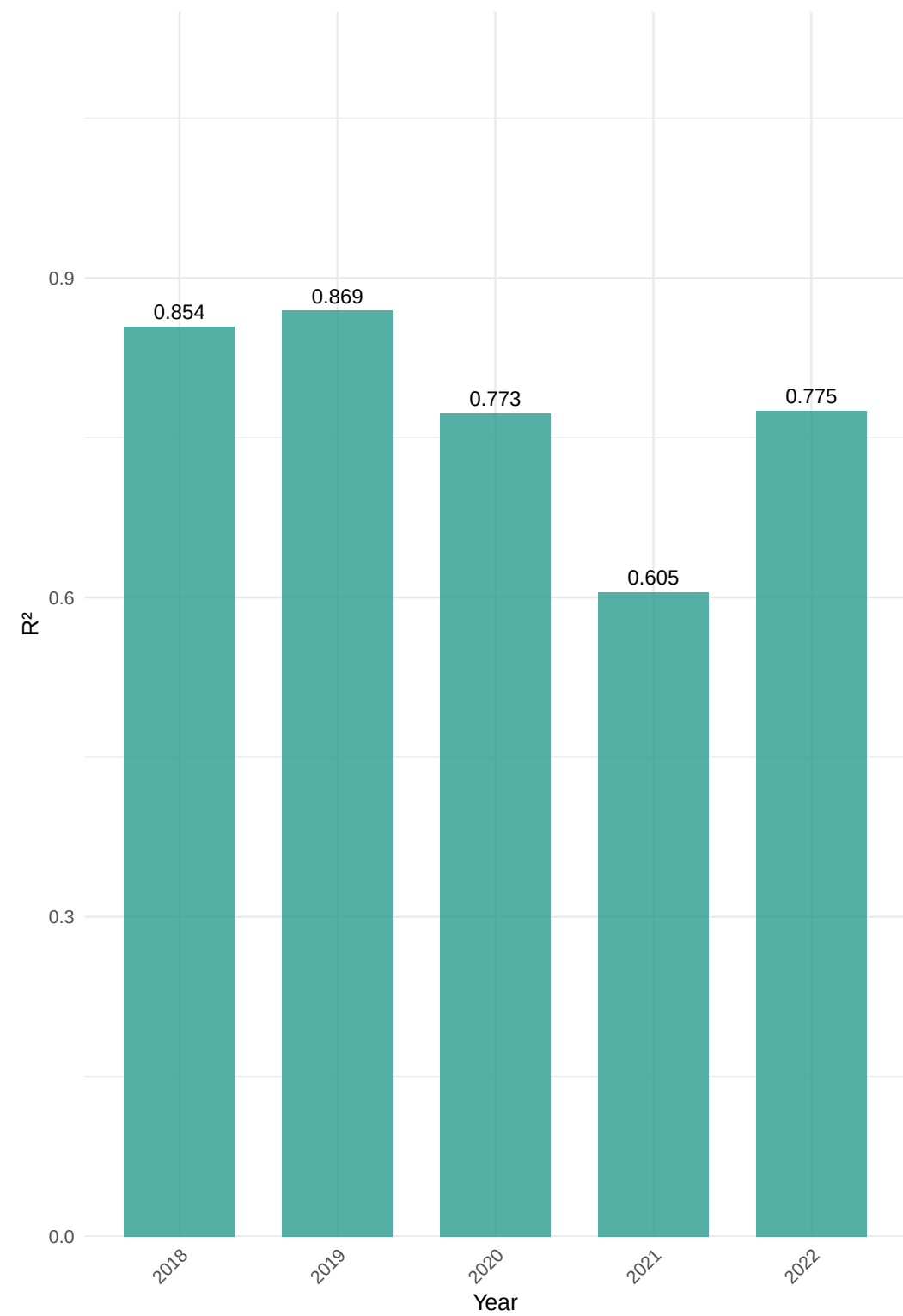


Performance Metrics by Year

MAE by Year



R² by Year



Sample Predictions – Year 2018 (Original Scale)

Lat: 31.51, Lon: -97.23

150
100
50
0

Lat: 33.94, Lon: -84.54

Acer Pollen (grains/m³)

6
4
2
0

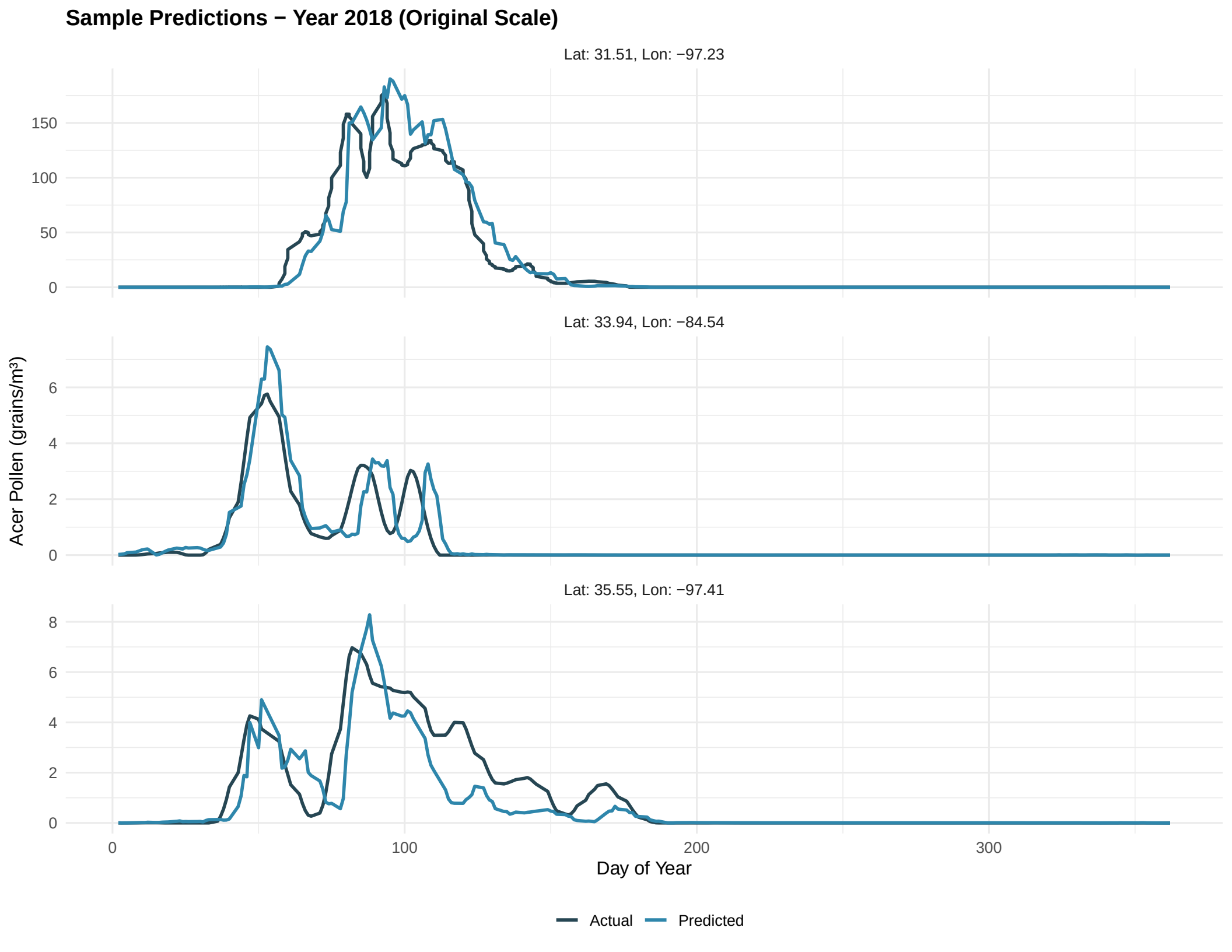
Lat: 35.55, Lon: -97.41

8
6
4
2
0

0 100 200 300

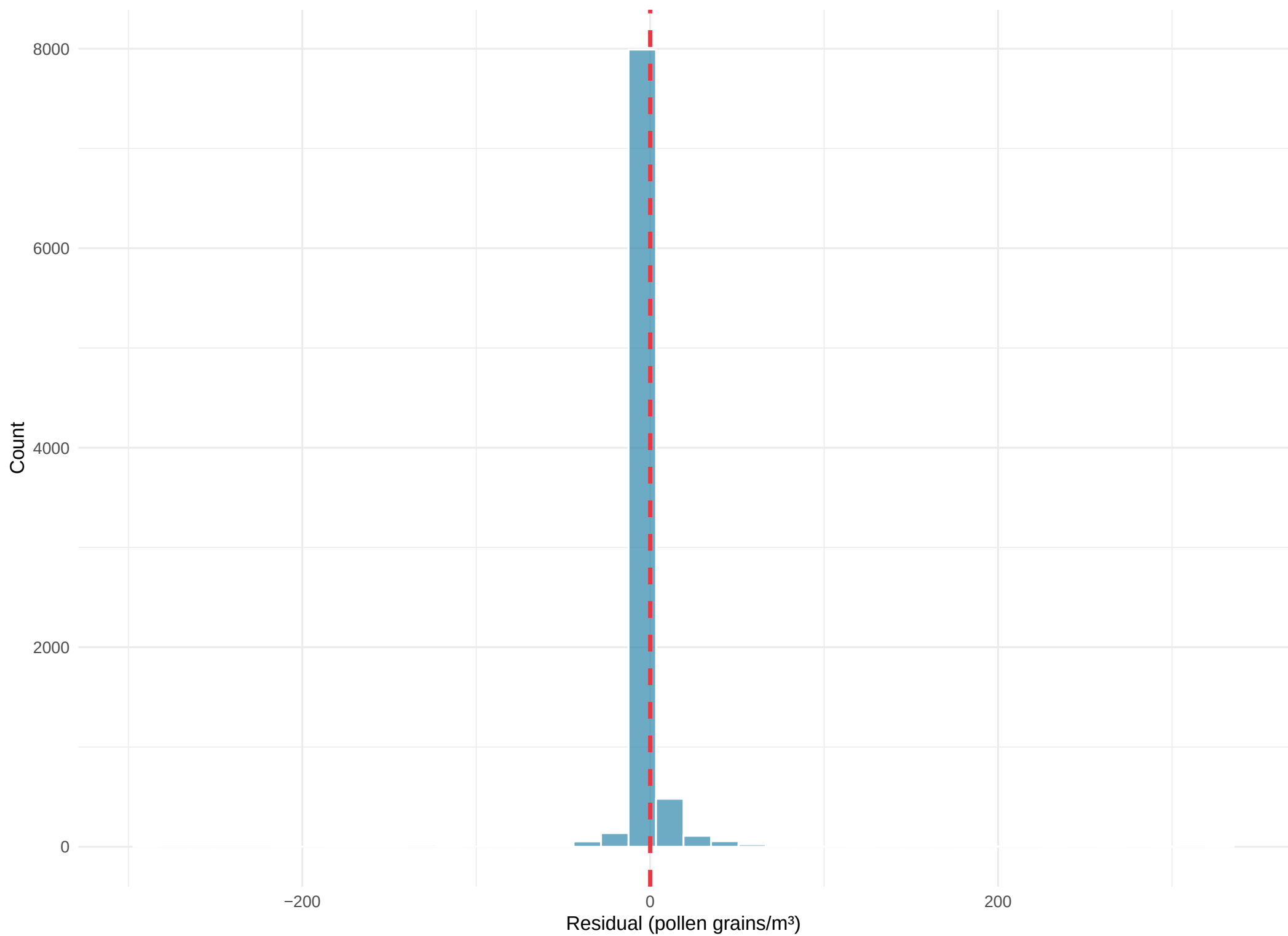
Day of Year

Actual Predicted

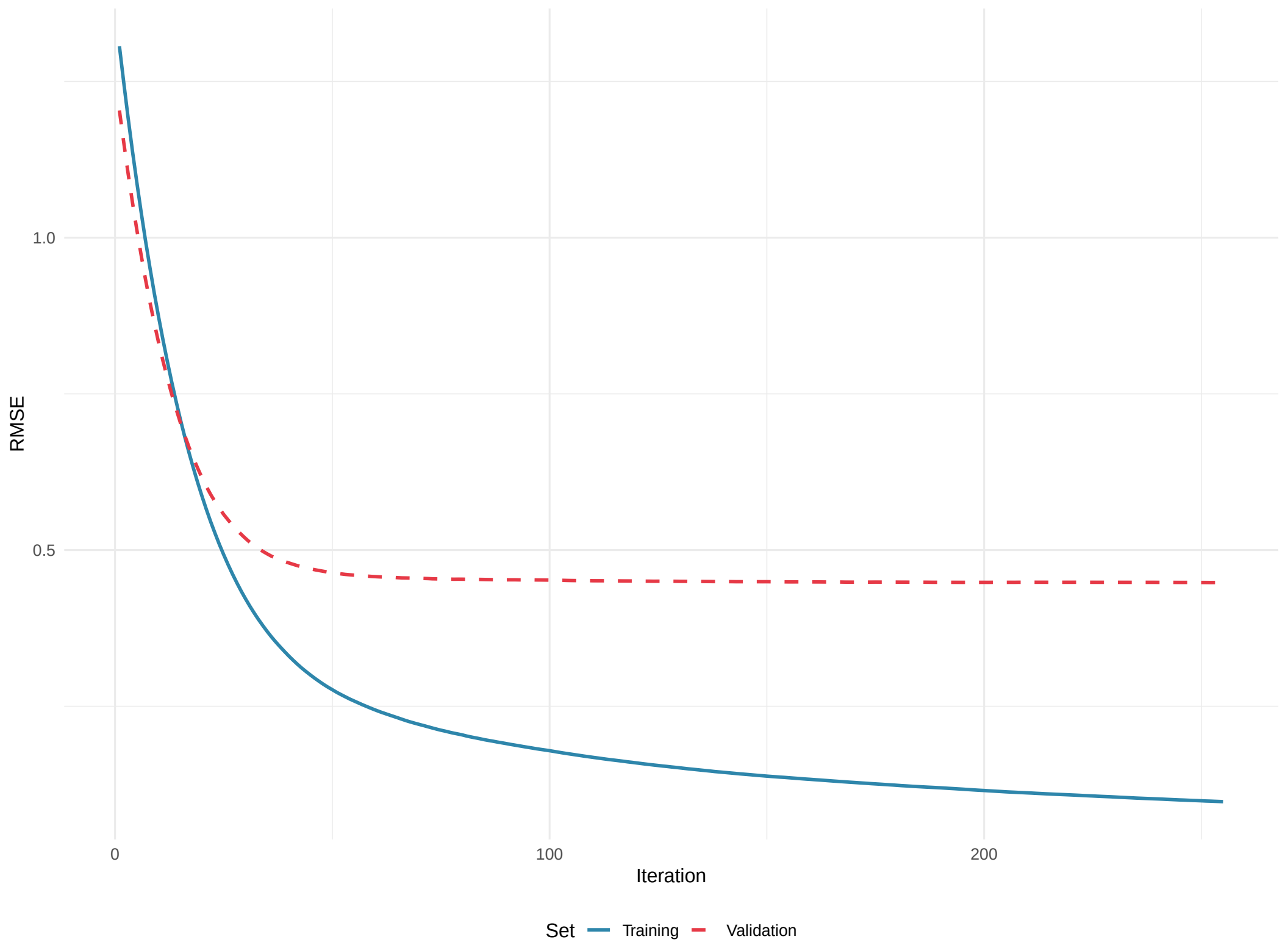


Residuals Distribution (Original Scale)

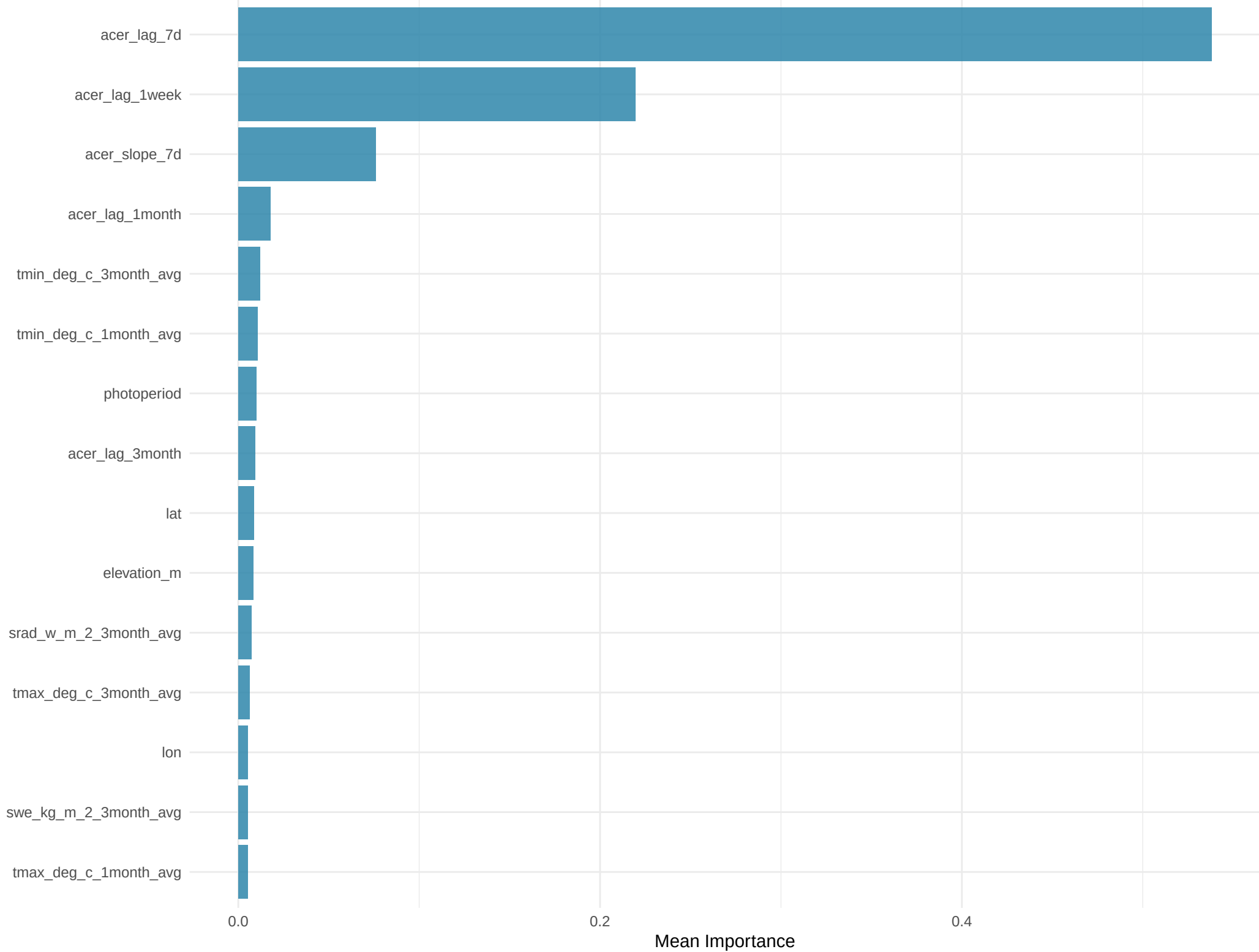
Mean: 0.81 | SD: 16.33 | Median: -0.00



Training vs Validation RMSE Curve



Top 15 Feature Importance (Avg Gain)



Summary Statistics

Test Set Performance (Original Scale)

RMSE: 16.34 pollen grains/m³

MAE: 3.67 pollen grains/m³

R²: 0.8159

Validation Performance

RMSE: 31.62 | MAE: 4.37 | R²: 0.6788

Duan Smearing Factor: 1.0025